

DOCUMENT : INVITATION PROPOSAL

AI Powered Autonomous Multi-Drone System for Smart
Agricultural Monitoring

TRINETRA DRONES AND ROBOTICS PVT.LTD

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Introduction

Modern agriculture faces challenges such as delayed crop disease detection, inefficient irrigation, pest infestations, and labor shortages. Traditional monitoring methods are time-consuming and often fail to provide timely insights. An intelligent drone-based monitoring system can help farmers make faster and more informed decisions.

Problem Statement

Manual crop inspection is labor-intensive and inefficient for large agricultural fields. Delayed identification of diseases, water stress, and pest attacks can lead to significant crop losses, increased production costs, and reduced agricultural productivity.

Proposed Solution

The proposed solution is an AI-powered autonomous drone system equipped with RGB and thermal cameras, GPS, and environmental sensors. The drone captures real-time field data, which is analyzed using Artificial Intelligence and Computer Vision to detect crop diseases, monitor irrigation needs, identify pest infestations, and assess overall crop health. The analyzed information is displayed on a cloud-based dashboard, enabling farmers to take timely corrective actions.

Technologies Used

- Autonomous Drone
- Artificial Intelligence (AI)
- Computer Vision
- Internet of Things (IoT)
- GPS
- RGB & Thermal Cameras
- Cloud Computing

System Workflow

Mission Planning → Autonomous Flight → Image & Sensor Data Collection → AI Analysis → Disease & Irrigation Detection → Cloud Dashboard → Farmer Alerts & Recommendations

Expected Benefits

- Early crop disease detection
- Improved irrigation management
- Reduced pesticide and fertilizer usage
- Higher crop yield and productivity
- Lower labor costs
- Sustainable farming practices

Future Scope

Future enhancements include drone swarm technology, weather prediction integration, edge AI for faster decision-making, renewable energy-powered drones, and satellite data integration for large-scale precision agriculture.

Conclusion

The proposed AI-powered drone system offers an efficient and sustainable solution for modern agriculture by enabling real-time monitoring, predictive analysis, and data-driven decision-making. It has the potential to improve crop productivity while reducing operational costs and supporting precision farming.